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IN THE RELATIONAL SANDBOX: DEEP DEMOCRACY AND TECHNOLOGY

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ABSTRACT

Designing for democracy often emphasizes values while overlooking the role of relationships in shaping civic life. This workshop explores how relationality – encompassing social ties, political agency, and economic conditions – shapes grassroots democratic experiments and the technologies they inspire. Grounded in principles of deep democracy and participatory approaches, we use the Relational Sandbox to conduct small-scale experiments on how democratic technologies can redistribute power and how locally rooted designs can resist extractive technology development. Through the lens of relational civics, we co-imagine strategies prioritizing meaningful, democratic technologies over capital-driven ones. The workshop invites designers, activists, and researchers to co-develop strategies and design guidelines that place relationships at the center of democratic practice.

INTRODUCTION

Design has always been relational, shaping and being shaped by the interplay between individuals, technologies, and social systems. While relationality has been explored across disciplines, recent design discourse has shifted beyond human-centered approaches, positioning relational design at the intersection of social, ecological, and technological systems (Akama et al., 2019; Escobar, 2018; Escobar, 2021; Helbing, 2021; Irwin et al., 2022; Nielsen & Bjerck, 2022). This shift challenges paradigms that prioritize human exceptionalism, instrumental rationality, and market-driven innovation, particularly in the face of growing corporate control over digital infrastructures and public life – which is also frequently reported in media outlets these days (Klein, 2020; Duffy, 2025; Kitchen, 2025). Instead, relational design emphasizes mutual dependencies, care, and collective responsibility for past, present, and future relations. It resonates with feminist and Indigenous epistemologies, which have long critiqued Western design traditions for their extractive and reductionist tendencies (Haraway, 2006). In the spirit of these traditions, this workshop challenges the direction of current technological development as a continuation of masculine power. Feminist scholarship has long done just that, emphasizing relationality and care, which are urgently needed in contemporary democracy, as well as interdependence as political practices (Tronto, 2020; de La Bellacasa, 2017), while critiquing the masculinist, extractive logics that underpin technological innovation and systems thinking (Wajcman, 1991; Suchman,

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2007). Activist design traditions, from critical making to design justice (Costanza-Chock, 2020), provide tools for resisting such logics by centering the needs, voices, and power of marginalized communities. In this light, our approach is critical not only of capital-driven and anti-democratic technologies but also of research cultures that reproduce these dynamics through depoliticized, technocratic, or market-oriented framings of technology design.

We challenge those tendencies by fostering spaces for relational civic experimentation, conceived as acts of collective imagination, framed as a *Relational Sandbox for Democracy* – a conceptual and experimental space for exploring democratic practices through the lens of relationships, including social, political, ecological, and technology mediated ones. The Relational Sandbox is not a fixed method but a flexible, participatory environment where one can engage in situated experiments, reflect on entanglements of power and care, and prototype technologies or interactions prioritizing relational values over extractive or transactional logic.

DEMOCRACY-TECHNOLOGY CRISIS

In turbulent times like the present, crises and frictions expose structural weaknesses in democratic systems, disrupt social norms, and create opportunities for more resilient and just alternatives (Gkeredakis et al., 2021). Calls for transparent, inclusive, and responsible technology have persisted for decades across various communities. Feenberg, one of the foremost critical technology theorists, wrote, *'The common-sense view of technology limits democracy to the state. By contrast, I believe that unless democracy can be extended beyond its traditional bounds into the technically mediated domains of social life, its use value will continue to decline, participation will wither, and the institutions we identify with a free society will gradually disappear'* (Feenberg, 1991). Today, that warning resonates more than ever. Feenberg also identified the non-democratic design of technology as one of the leading causes of the erosion of democracy, "*… the degradation of labour, education, and environment is rooted not in technology per se but in antidemocratic values that govern technological development"* (Feenberg, 2002). This concern is even more pressing with the rise of opaque AI-based technologies. Sclove similarly highlights the importance of more democratic politics of technology, '*Among the panoply of questions concerning technology that escape attention, perhaps the most important one involves how technology bears on democracy'* (Sclove, 1995, p.8).

These concerns align with other influential thinkers. Latour (Latour, 2007; Latour & Porter, 1996) asserts that material and technological mediators profoundly shape social and political life, while Dewey (2016) calls for a participatory democracy rooted in active civic engagement. As Bjørkdahl (2023) elaborates, Dewey's work highlights the need for inclusive, deliberative processes that empower citizens and adapt democratic principles to evolving societal contexts – which increasingly involve technology.

Taken together, these thinkers emphasize the transformative potential of technology in addressing the contemporary challenges faced by democracy. Yet, their voices have not led to meaningful structural changes in technology design practices. Still, this is not a time for resignation; rather, it is a time to better use our tools to explore alternatives. This workshop is offering a space to explore how relational approaches to design can help realize the democratic potential of technology.

DESIGN AND DEEP DEMOCRACY

Participatory design (PD), especially its reimagined form that focuses on big social challenges (Bannon et al., 2018; Bødker & Kyng, 2018; Bødker et al., 2022), offers a promising design trajectory. It aligns well with Feenberg's vision of democratic technology design. Although power issues have been addressed in PD, e.g. (Bratteteig & Wagner, 2014), today's challenges are more complex. With its increased focus on social good, PD still has much to offer for addressing the entanglement of design, technology, and democracy.

Recently, platforms supporting engagement with civics have emerged, like vTaiwan (Hsiao et al., 2018) and communing initiatives (Bollier & Helfrich, 2015) aimed at creating more democratic, accountable, and transparent systems for participation, engagement, and community building. However, designing for complexity demands systemic approaches.

Transition design (Bisson et al., 2022; Irwin et al., 2022; Escobar, 2018) offers a future-oriented, sustainable framework for systems thinking. In particular, tools and techniques from designing for complexity (Sevaldson, 2022), such as Giga mapping, will be used in the workshop to discuss relationships. Therefore, we draw on PD and transition design as the theoretical backdrop for our workshop. We utilize their toolboxes, including the ideas behind the plurishop workshop format (Culén et al., 2023; Junge & Culén, 2023) to create our sandbox for supporting technology design for deep democracy.

Deep democracy, as conceptualized by Green (1999), implies a set of values to build upon, an aspirational and desirable new form of empowerment and transformation of democratic culture. Deep democracy also draws on Dewey's work. It promotes democracy as a way of life that moves beyond tensions, such as individual-community, silence-activism, or inclusion-exclusion. It calls for relational care and thoughtful engagement.

WORKSHOP DETAILS

Through this workshop, facilitated by an interdisciplinary team of researchers and designers, various perspectives on relationality, power, technology, and deep democracy will be explored. Our goal is to better understand strategies compatible with deep democracy and the importance of relationships in designing future civics technologies. To that end, we use a Relational Sandbox, explained in more detail below.

RELATIONAL SANDBOX

The main design material used in the workshop is the dynamics of relationships - including power and care - in designing technologies that better support democratic values. Civic structures are traditionally complex and resistant to change. The Relational Sandbox offers a space to question traditional dynamics and engage in collective sense-making around more just and socially sustainable alternatives, bridging deep democracy, relational design, and participatory experimentation.

As stated in the introduction, a Relational Sandbox is a space for imaginative and participatory experimentation with civic relations. Inspired by the use of sandboxes in fields like psychology, innovation, and policy (Leckenby et al., 2021; Allen, 2019), we adapt the concept to create a risk-free, exploratory environment for flexible and expressive thinking. Just as traditional sandboxes allow developers to model and test systems before implementing them, the Relational Sandbox enables participants to focus on relationships while thinking creatively, prototyping new technological alternatives, and critically reflecting on their implications.

WORKSHOP ACTIVITIES

We envision activities split into three parts: 1) relational mapping, 2) power dynamics prototyping, and 3) co-developing relational design strategies for engaging with deep democracy.

We kindly ask participants to *prepare in advance*. Participants are encouraged to reflect on grassroots democratic initiatives involving technology, such as platforms or social media. Are they novel or adapted from existing concepts? Do they pursue specific democratic goals? Who is involved, and what values do they support? Considering local initiatives in advance will support more active participation.

The workshop will begin with a short motivation and a round of introductions, including a brief mention of a local initiative by each participant. These examples will serve as starting points for the workshop activities.

Participants will form three groups, each working on a different type of initiative.

1. *Relational mapping and civic networks* (30 min)

Goal: Identify relationships surrounding the initiative and examine how they shape civic engagement, including how technology supports or hinders that engagement. The Relational Sandbox approach allows for dynamic rather than static exploration of these ties.

Activity: Creating collective civic maps. Teams create a visual map of civic relationships (e.g., interactions in activism, governance, community, or tech projects). These maps showing collective networks will help identify power structures, gaps, technological influences, and shifting relationships.

Discussion prompts: Where does power flow? Who is excluded? What role does technology play?

2. *Power dynamics prototyping* (50 min)

Goal: Experiment with relational models of governance and civic participation.

Activity: Role-play: the power of technology in design for democracy. Participants will take on roles (e.g., grassroots organizer, policymaker, tech executive, designer, or (marginalized) community member) within the initiative. They will enact several configurations of the relationships, negotiating governance structures and power dynamics by simulating them and testing different options**.**

Discussion: How do relationships shift power? What alternative governance structures emerge?

3. *Relational design for deep democracy* (50 min)

Goal: Explore co-developed methodologies and design guidelines for care relationships.

Activity: Futures prototyping. Based on previous discussions, teams will sketch or design a relational civic tech intervention. We are particularly attentive to the emergence of care, compassion, and empathy as possible counterpoints to the dominance of commercial technologies and profit-driven logic. Thus, relational models that facilitate alternative ways of imaging technologies for democratic initiatives should emerge. Participants may optionally use AI tools (e.g. ChatGPT or image generation) to support their designs.

Discussion: Presentations, followed b collective synthesis.

Closing reflection (15 min)

Facilitated discussion: How do these insights shift the way we approach technology design?
Follow-up: We recognize that post-workshops engagement is often a limitation. To address this, we will actively support continued conversations, reflections and collaborations beyond the event. Participants will have the option to build on this work

through research, design, methods, methodologies, or technology development.

Table 1: The Schedule Overview with Key Activities

Duration	*Activity*
15 min.	Introduction. Motivating the WS and a brief introduction of WS participants
30 min.	Relational mapping and civic networks
5 min.	Break
50 min.	Power dynamics prototyping
10 min.	Break
50 min.	Relational design for deep democracy
5 min.	Break
15 min.	Closing reflections

CONCLUSION

The activities directly support help the goals of the workshop, benefiting all present. Participants should experience relational design and its connection to democracy and have an opportunity to develop a network of like-minded designers, researchers, and educators for potential future collaboration. The organizers will also expand their collaborative network and gain practical experience and deeper understanding of using Relational Sandboxes to support the development of more democratic design methodologies, guidelines, and outcomes.

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